

1. M/J/23/21/Q1

An aircraft pulls a glider along a runway as shown in Fig. 1.1.

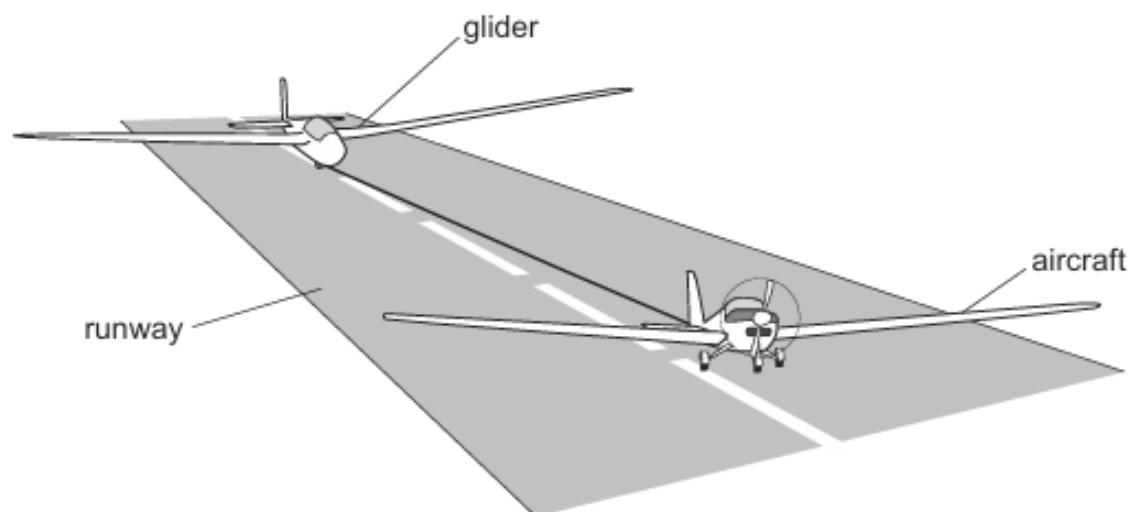


Fig. 1.1

Fig. 1.2 shows the speed of the glider during the first 12 s of the motion.

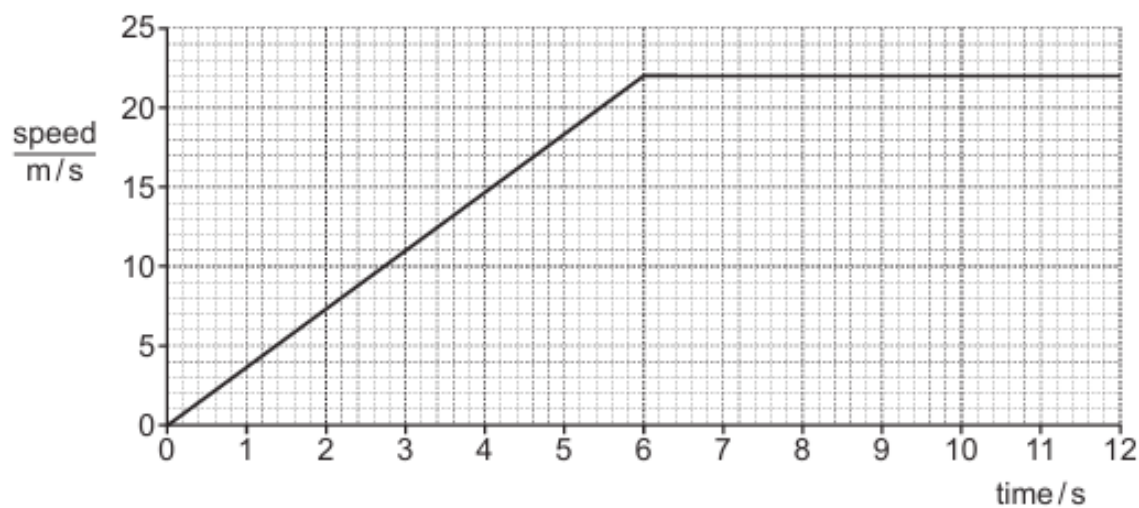


Fig. 1.2

(a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = m [2]

2. M/J/23/22/Q1

Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

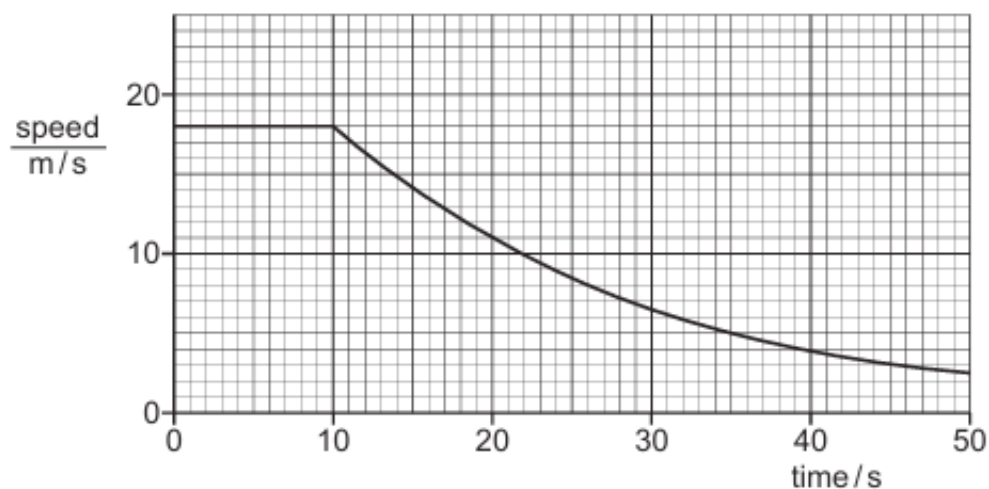


Fig. 1.1

- (a) Describe the motion of the car shown in Fig. 1.1.

.....

.....

.....

..... [2]

- (b) At time $t = 10$ s the engine of the car is switched off. The brakes are not applied.

- (i) Name **two** forces that act on the car to cause the change in motion after $t = 10$ s.

1

2

[1]

- (ii) Suggest why Fig. 1.1 is a curve after $t = 10$ s.

.....

.....

..... [1]

3. O/N/22/21/Q7

Fig. 7.1 shows a child sitting on a sledge on a snow-covered hill of constant slope.

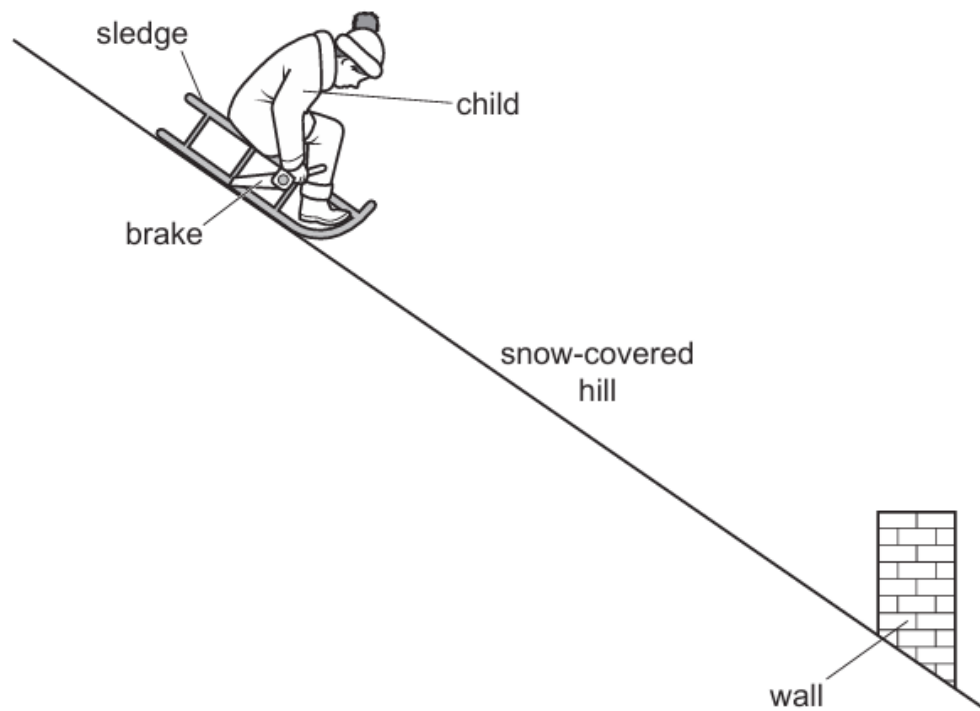


Fig. 7.1 (not to scale)

At time $t = 0$, the child and the sledge begin to move down the hill in a straight line.

When the child sees a wall ahead, he applies the brake.

The child and sledge continue to travel in a straight line until they come to a stop before hitting the wall.

Fig. 7.2 is the speed-time graph for the journey.

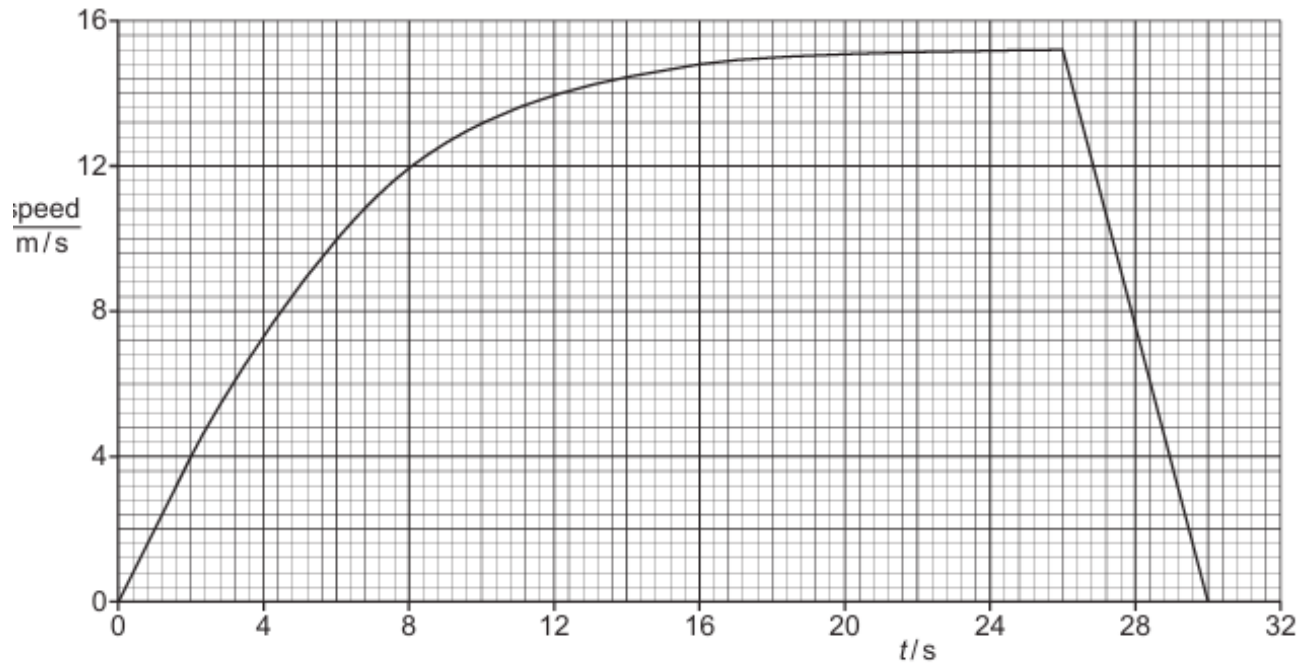


Fig. 7.2

The brake is applied at $t = 26$ s.

(a) Fig 7.2 shows how the speed of the child and sledge varies over the whole of the journey.

Explain why, between $t = 0$ and $t = 26$ s, the speed varies in the way shown by the graph.

.....

.....

.....

.....

.....

..... [3]

(b) At $t = 26$ s, the front of the sledge is 35 m from the wall.

Determine the distance between the front of the sledge and the wall when the sledge stops.

distance = [3]

(c) At $t = 26\text{ s}$, the child and sledge begin to decelerate.

(i) Determine the size of the deceleration.

deceleration = [3]

(ii) The mass of the child is 46 kg and the mass of the sledge is 9.0 kg .

Calculate the resultant force on the child and sledge as they decelerate.

resultant force = [2]

(iii) State the energy transfer that is taking place as the child and sledge decelerate.

.....
.....
..... [2]

(d) At $t = 26\text{ s}$, when the brake is first applied, the child jerks forwards on the sledge.

Explain why.

.....
.....
..... [2]

[Total: 15]

4. O/N/22/22/Q1

A train travels along a straight horizontal track. At time $t = 0$, the train passes through station P at constant speed without stopping.

The driver applies the brakes 70 s before reaching station Q. The train decelerates.

Fig. 1.1 is the speed–time graph for the train from $t = 0$ until it stops at station Q.

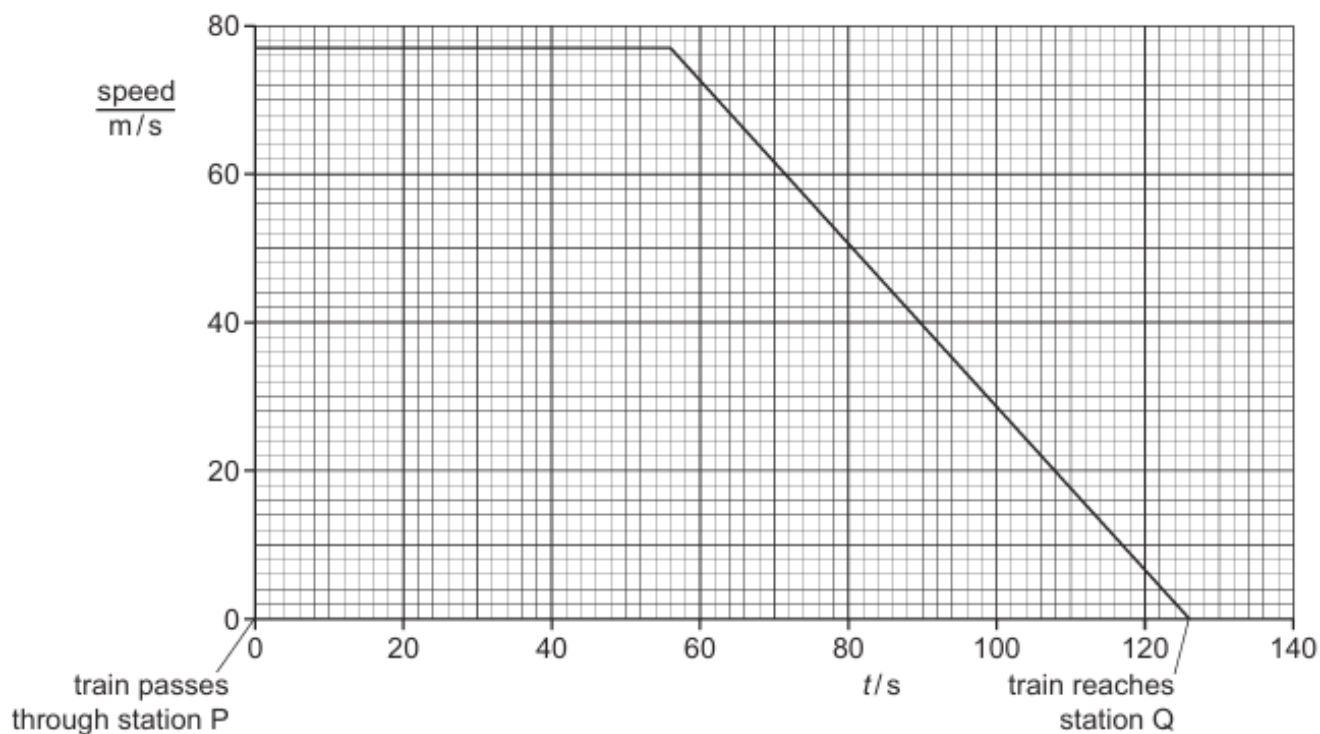


Fig. 1.1

(a) Using Fig. 1.1, determine the distance between station P and station Q.

distance = [3]

(b) The mass of the train is $3.8 \times 10^5 \text{ kg}$.

(i) Determine the deceleration of the train in the 70 s before it stops at station Q.

deceleration = [2]

(ii) Calculate the resultant force on the train as it decelerates.

resultant force = [2]

[Total: 7]

5. M/J/22/22/Q7

Fig. 7.1 shows the speed–time graph for a car travelling on a straight horizontal road.

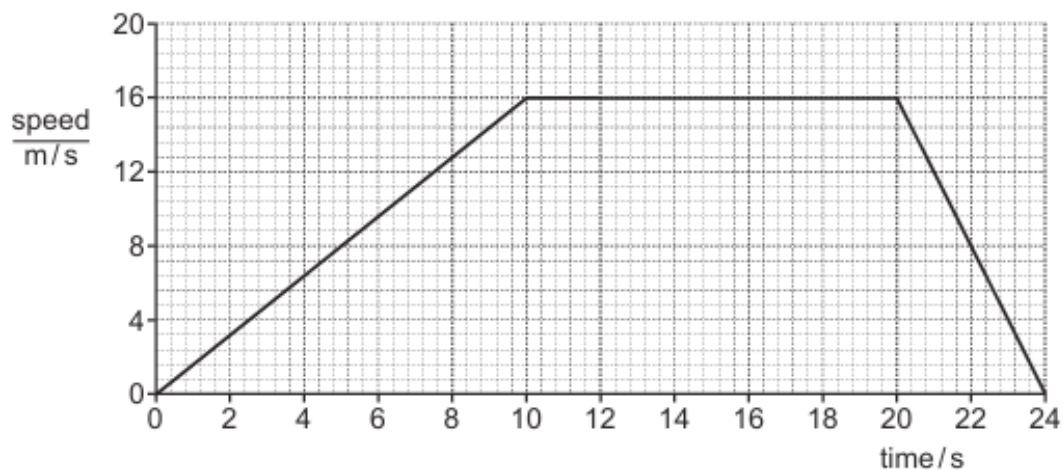


Fig. 7.1

- (a) (i) Describe the motion of the car.

.....

.....

.....

.....

..... [3]

- (ii) Using Fig. 7.1, calculate the distance travelled by the car during the 24 s of its motion.

Show your working.

distance = [3]

- (iii) Calculate the average speed of the car during its motion.

average speed = [2]

- (iv) A second car travels at a steady speed. It travels the same distance as the first car in the 24 s of the journey.

On Fig. 7.1, draw the speed–time graph for the second car.

[2]

6. M/J/21/21/Q1

An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

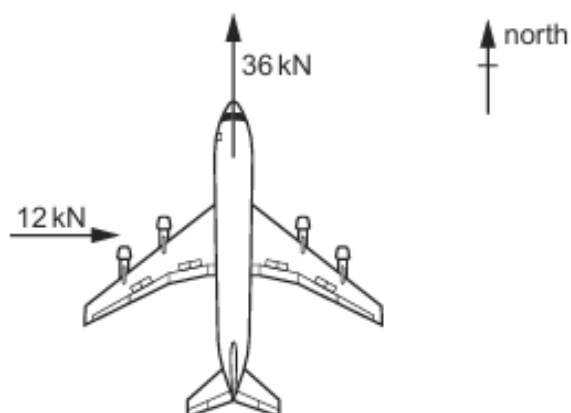


Fig. 1.1 (not to scale)

- (a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force =

angle =

[3]

(b) The acceleration of the aircraft is uniform.

(i) Describe how a uniform acceleration differs from a non-uniform acceleration.

.....
.....
.....
..... [2]

(ii) The mass of the aircraft is 60 000 kg.

Calculate the acceleration of the aircraft.

acceleration = [2]

[Total: 7]

7. M/J/21/22/Q2

A car approaches a set of traffic lights. The lights change to red at time $t = 0$.

Fig. 2.1 shows how the speed of the car changes with time.

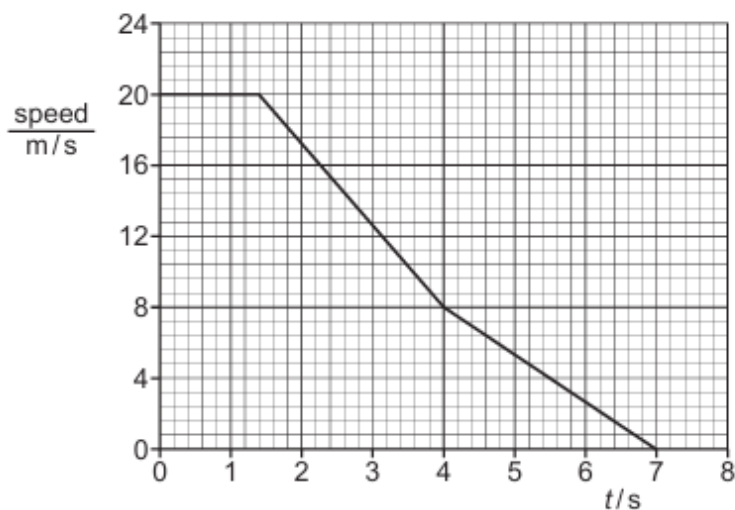


Fig. 2.1

- (a) The car starts to slow down a short time after the lights change to red.

Determine the time between the lights changing to red and the car starting to slow down.

..... [1]

- (b) (i) State what is meant by *uniform acceleration*.

.....
 [1]

- (ii) State how Fig. 2.1 shows that the deceleration of the car between $t = 2\text{ s}$ and $t = 7\text{ s}$ is *non-uniform*.

.....
 [1]

- (c) Determine the distance the car travels from the moment the car starts to slow down until it stops.

distance = [3]

[Total: 6]

8. O/N/20/21/Q1

Fig. 1.1 is the distance–time graph for a skydiver who jumps from a balloon at time $t = 0$.

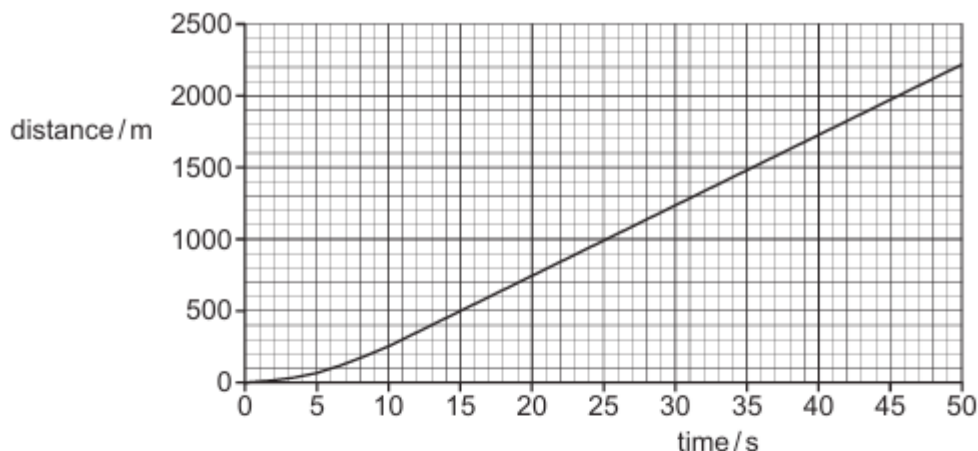


Fig. 1.1

- (a) The first part of the graph shows the motion of the skydiver from when he jumps until he reaches terminal velocity.

- (i) Describe the motion of the skydiver between $t = 0$ and $t = 20$ s.

.....

 [2]

- (ii) Explain the motion of the skydiver between $t = 0$ and $t = 20$ s in terms of the forces acting on him.

.....

 [3]

- (b) Using Fig. 1.1, determine the terminal velocity of the skydiver.

On Fig. 1.1, indicate any values used for your calculation.

terminal velocity = [3]

[Total: 8]

9. O/N/20/22/Q7

A bus leaves a bus-stop at time $t = 0$ and travels along a horizontal road until it reaches a second bus-stop. Fig. 7.1 is the distance-time graph for the bus between $t = 0$ and $t = 60$ s.

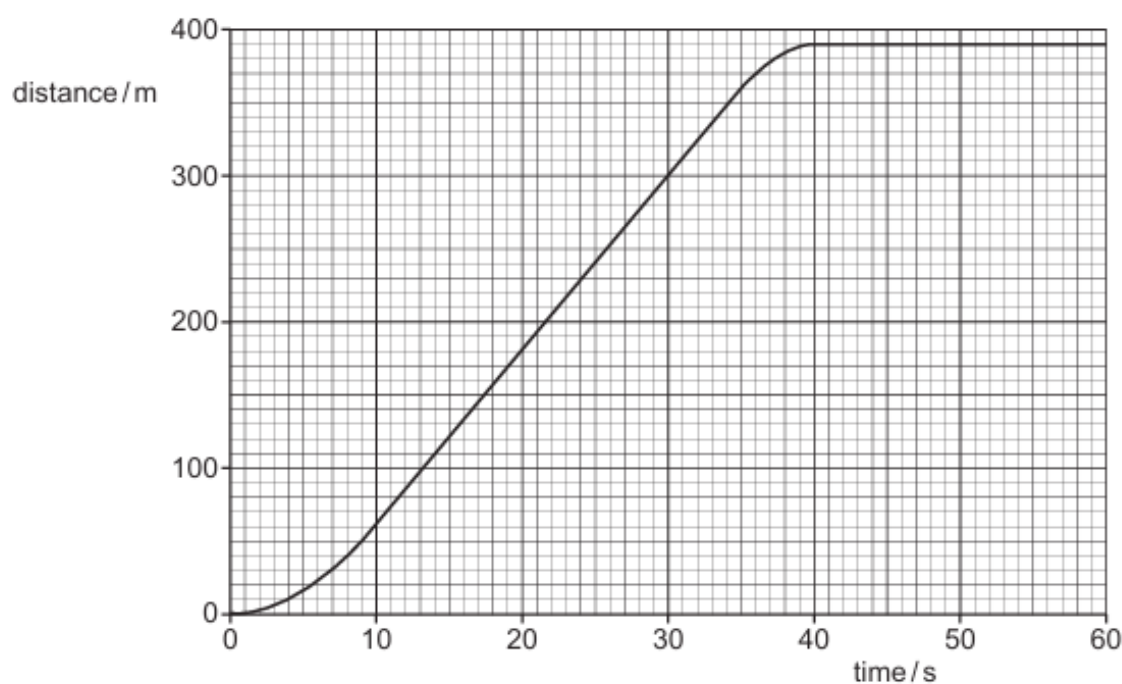


Fig. 7.1

The road on which the bus is travelling is straight except for a short, curved section. The bus travels around this circular curve between $t = 21$ s and $t = 24$ s.

- (a) Describe how the motion of the bus between $t = 0$ and $t = 10$ s differs from its motion between $t = 35$ s and $t = 40$ s.

.....

.....

.....

.....

..... [3]

(b) Determine:

(i) the maximum speed of the bus during these 60 s

maximum speed = [3]

(ii) the average speed of the bus between leaving the first bus-stop and arriving at the second bus-stop.

average speed = [2]

10. M/J/20/22/Q1

Fig. 1.1 shows part of the speed–time graph for an athlete in a race.

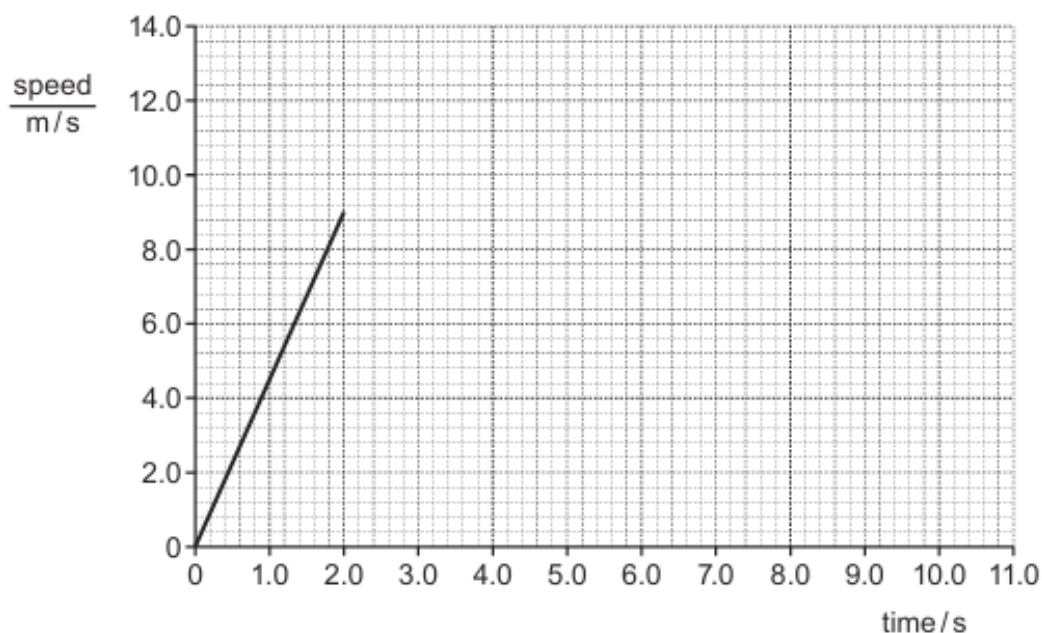


Fig. 1.1

- (a) During the race, the acceleration of the athlete is uniform in the first 2.0 s.

State how the graph shows that the acceleration is uniform.

.....
 [1]

- (b) Determine the distance travelled by the athlete in the first 2.0 s.

distance = [2]

- (c) During the rest of the race:

- from 2.0 s to 5.5 s, the acceleration of the athlete decreases
- at 5.5 s, the athlete reaches a maximum speed of 12 m/s
- from 5.5 s to 8.0 s, the athlete travels at a speed of 12 m/s
- from 8.0 s to 11.0 s, the athlete decelerates, finishing the race at a speed of 10 m/s.

On Fig. 1.1, complete the speed–time graph for times between 2.0 and 11.0 s. [3]

[Total: 6]

11. O/N/19/22/Q1

There is no atmosphere on the Moon.

An astronaut on the Moon drops a feather and a hammer from the same height at the same time. They both accelerate downwards at 1.6 m/s^2 and they hit the ground at the same time.

(a) The weight of the hammer is much larger than that of the feather.

Explain, in terms of their weights and masses, why their accelerations are equal.

.....

.....

.....

.....

.....

..... [3]

(b) Both the feather and the hammer take 1.5 s to fall to the ground from rest.

(i) Calculate the speed of the objects as they hit the ground.

speed = [2]

(ii) On Fig. 1.1, draw the speed-time graph for the fall. At the correct position on the y-axis, write the value of the speed at time $t = 1.5$ s.

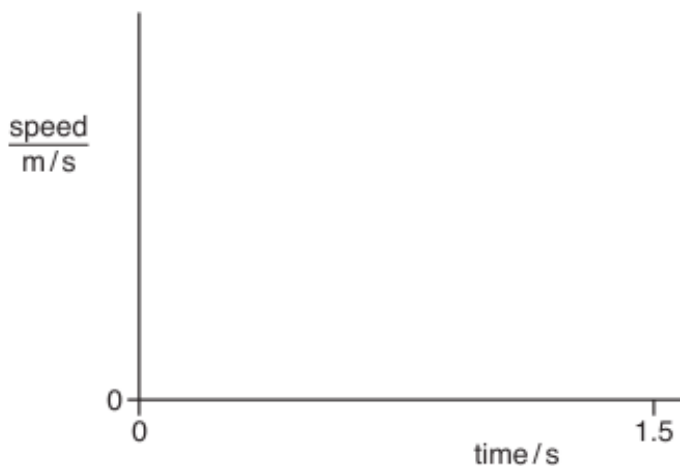


Fig. 1.1

[1]

(iii) Using the speed-time graph in (b)(ii), determine the height from which the objects are dropped.

height = [2]

[Total: 8]

12. M/J/19/21/Q1

Fig. 1.1 shows the distance-time graph for a journey made by a cyclist between town A and town B.

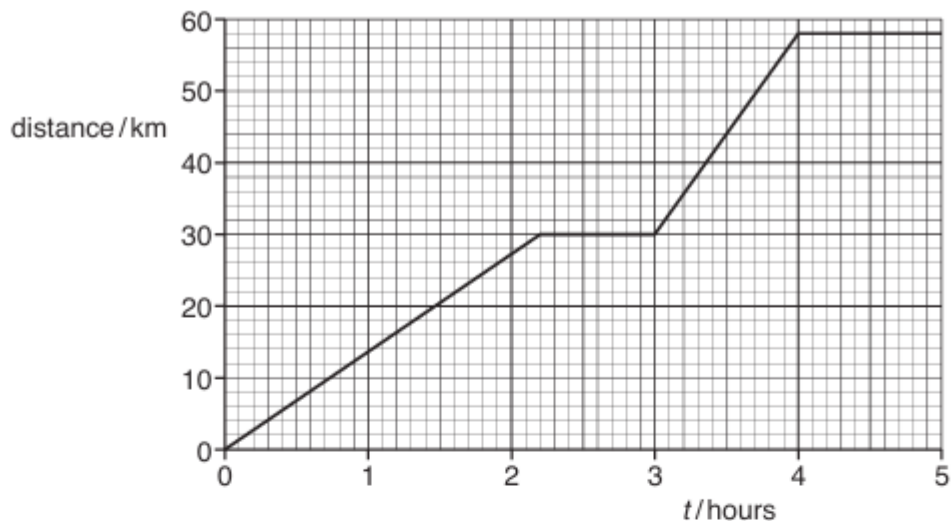


Fig. 1.1

The cyclist leaves town A at time $t = 0$ and arrives at town B at $t = 4.0$ hours.

- (a) (i) Determine the distance between the two towns.

distance = [1]

- (ii) Determine the time for which the cyclist rests during the journey, before he reaches town B.

time = [1]

- (b) The speed of the cyclist near the end of the journey is greater than the speed at the beginning.

State how the graph shows this.

.....
 [1]

- (c) Calculate the average speed of the cyclist for the journey from A to B.

average speed = [2]

[Total: 5]

13. M/J/19/22/Q9

Fig. 9.1 shows a map of the route taken by a car as it travels from town A to town B.

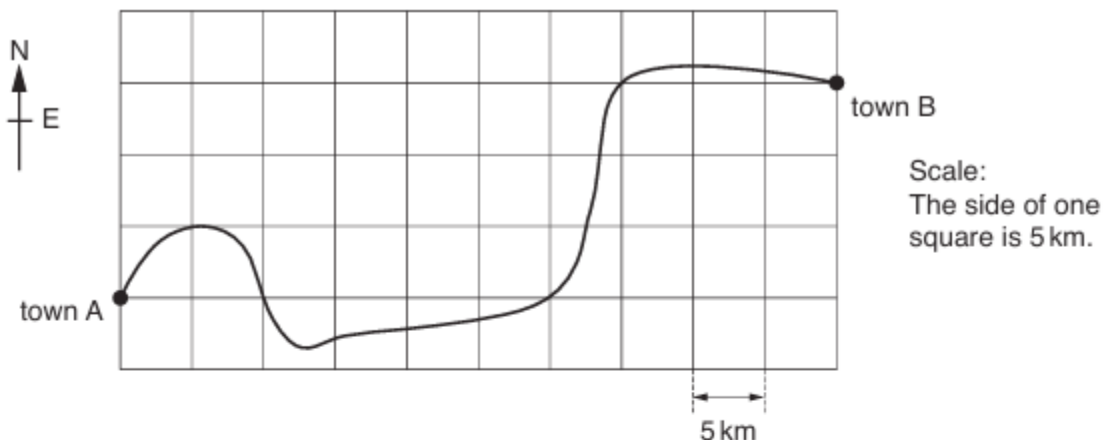


Fig. 9.1

The journey involves a distance and a displacement.

(a) (i) Using the scale on Fig. 9.1, determine the displacement in travelling from A to B.

size of displacement =

direction of displacement =

[3]

(ii) State how *distance* differs from *displacement*.

.....

 [1]

(iii) In some parts of the journey, the car is travelling at a constant speed but is also accelerating.

Explain how this is possible.

.....

 [2]

14. M/J/18/22/Q1

A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

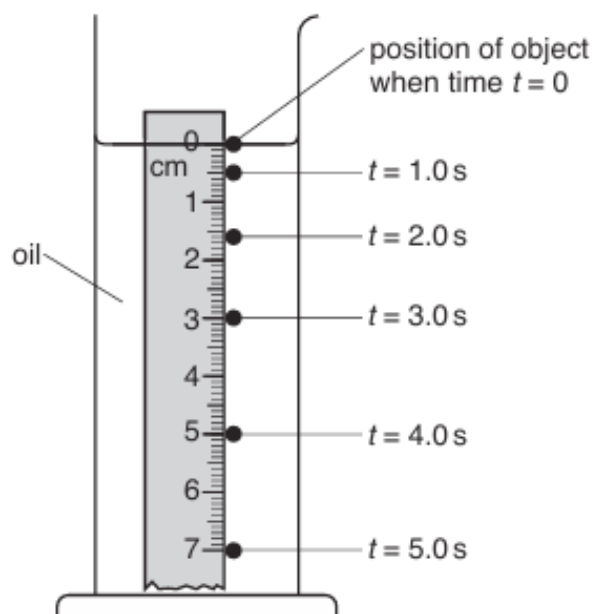


Fig. 1.1

(a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

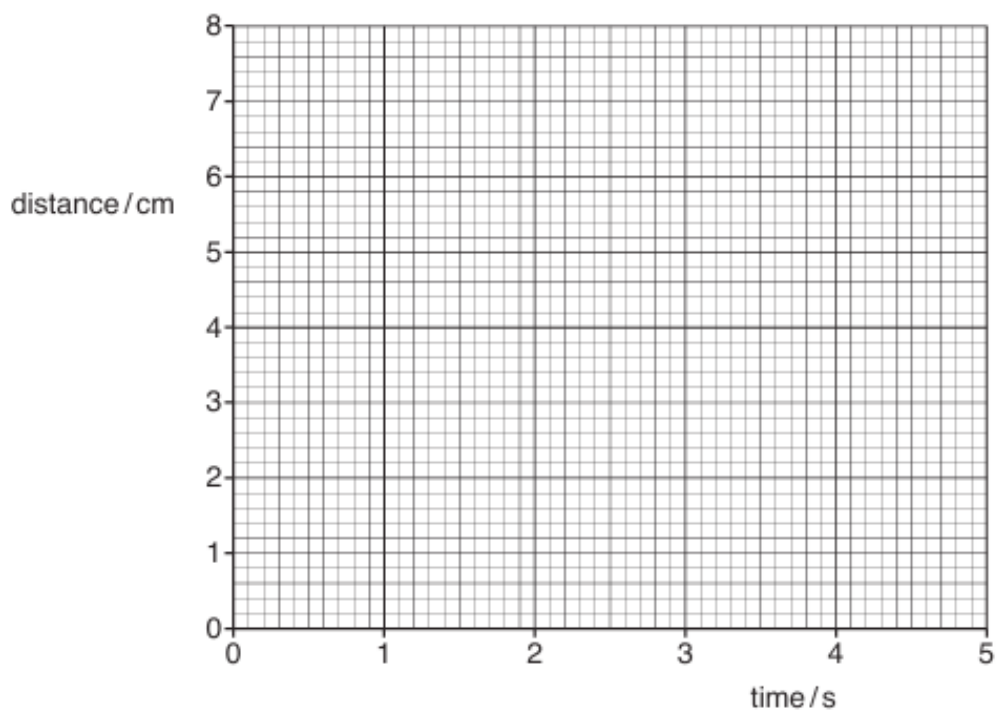


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....

.....

.....

.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....

.....

.....

.....[2]

15. M/J/18/21/Q1

A cyclist travels along a straight, horizontal road, as shown in Fig. 1.1.



Fig. 1.1

- (a) Describe how the average speed of the cyclist can be measured.

In your account, suggest what equipment is used, the measurements that are made and how the average speed is calculated.

.....

.....

.....

.....

.....

.....[3]

- (b) For part of the journey, the cyclist has a constant, negative acceleration.

- (i) State what is meant by a *negative acceleration*.

.....

.....[1]

- (ii) The size of the acceleration is 1.8 m/s^2 .

Calculate the change in speed of the cyclist in a time of 2.5 s.

change in speed =[2]

16. O/N/18/21/Q1

A girl of mass 35 kg, on a bicycle, accelerates from rest and travels down a slope in a straight line. The girl does not use the pedals.

Fig. 1.1 shows that the gradient of the slope is constant.

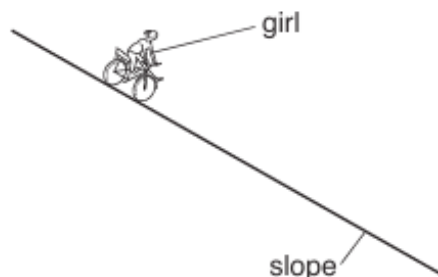


Fig. 1.1

- (a) Calculate the resultant force on the girl when she is accelerating at 2.6 m/s^2 .

resultant force = [2]

- (b) At first, her acceleration is constant. At time t_1 , her acceleration starts to decrease gradually until she is travelling at a constant speed in a straight line.

- (i) On Fig. 1.2, sketch a speed-time graph for the girl from when she starts moving until she is travelling at a constant speed. [2]

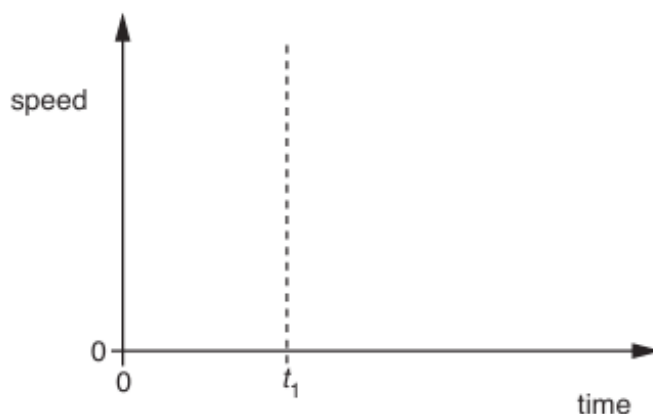


Fig. 1.2

- (ii) State how the distance travelled is found from a speed-time graph.

.....
 [1]

17. M/J/17/22/Q1

A car accelerates from rest in a straight line. During the first 14 s, the acceleration is uniform and the car reaches a speed of 25 m/s.

- (a) (i) Calculate the acceleration of the car.

acceleration =[2]

- (ii) After the first 14 s, the speed of the car continues to increase but the acceleration decreases. From 70 s to 80 s after the start, the car moves at a constant speed of 55 m/s.

On Fig. 1.1, draw a possible speed-time graph for the car.

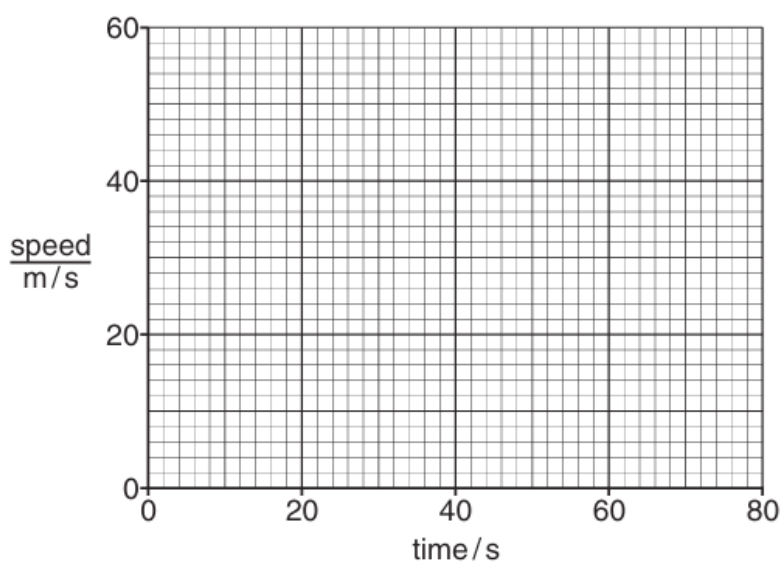


Fig. 1.1

[2]

18. O/N/16/22/Q1

A skier sets off from rest and accelerates uniformly at 3.4 m/s^2 in a straight line for 5.0 s.

- (a) Calculate the speed of the skier after 5.0 s.

speed =[2]

- (b) At 5.0 s, the skier stops accelerating and travels on for a further 10.0 s at a constant speed.

- (i) State the size of the resultant force acting on the skier during these 10.0 s.

.....[1]

- (ii) On Fig. 1.1, sketch a speed-time graph for the skier during the whole 15.0 s. [3]

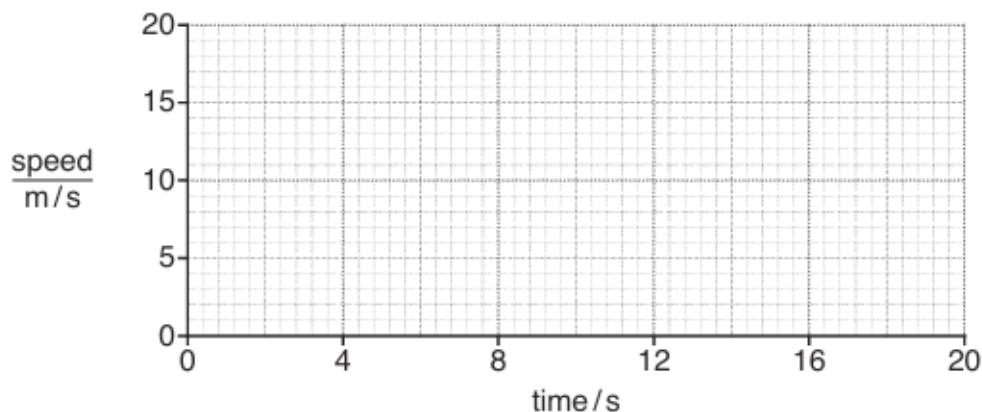


Fig. 1.1

- (iii) State how the distance travelled by the skier can be determined using the speed-time graph.

.....
.....[1]

19. O/N/15/21/Q1

Fig. 1.1 shows an archer firing an arrow at a target during a competition.

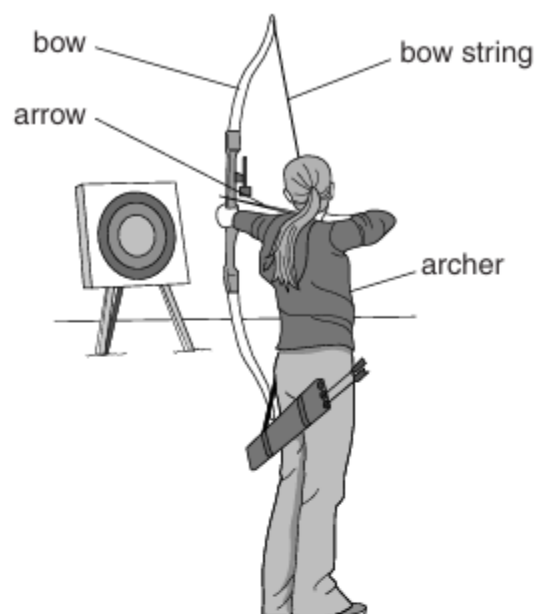


Fig. 1.1

The arrow has a mass of 0.018 kg and is initially at rest. The arrow leaves the bow 0.011 s after the bow string is released. When it leaves the bow, the velocity of the arrow is 95 m/s .

(a) Calculate

(i) the average acceleration of the arrow during the 0.011 s ,

average acceleration = [2]

(ii) the average force exerted on the arrow during the 0.011 s .

average force = [2]

(b) In practice, the force exerted on the arrow gradually decreases to zero during firing.

On Fig. 1.2, sketch a possible speed-time graph for the arrow as it is being fired.

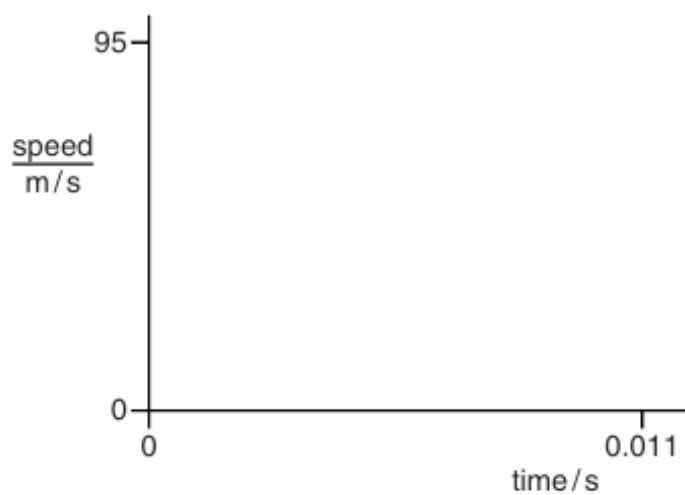


Fig. 1.2

[2]

20. M/J/15/21/Q1

Fig. 1.1 shows the distance-time graph for two cyclists A and B. They start a 500 m race together but finish the race at different times.

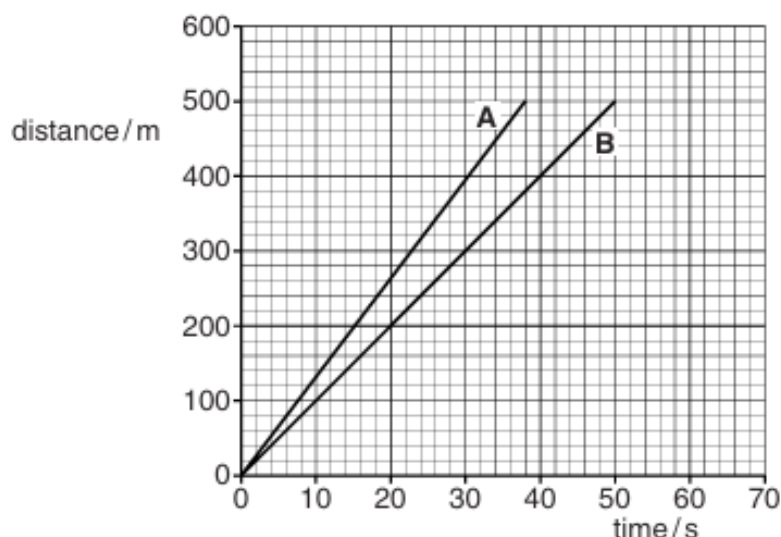


Fig. 1.1

(a) Use Fig. 1.1 to determine

(i) the distance between A and B at time $t = 20$ s,

.....[1]

(ii) the difference in the time taken by A and B for the race.

.....[1]

(b) Cyclist C starts the race at the same time as A and B and covers the first 200 m of the race at a constant speed of 5.0 m/s. He then accelerates and finishes the race at $t = 60$ s.

(i) On Fig. 1.1, draw the distance-time graph for cyclist C.

[2]

(ii) Calculate the average speed of cyclist C for the whole race.

speed =[2]

21. M/J/15/22/Q11

A bungee jumper falls from a bridge above a river, as shown in Fig. 11.1.

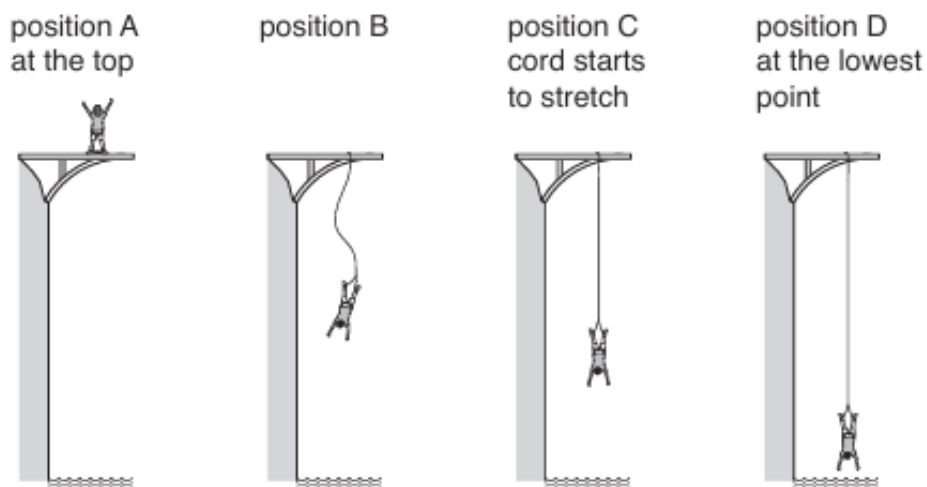


Fig. 11.1 (not to scale)

The man starts from position A in Fig. 11.1. The elastic cord starts to stretch at position C and he stops for the first time at position D. He continues to rise and fall.

Fig. 11.2 shows how the velocity of the man varies with time t .

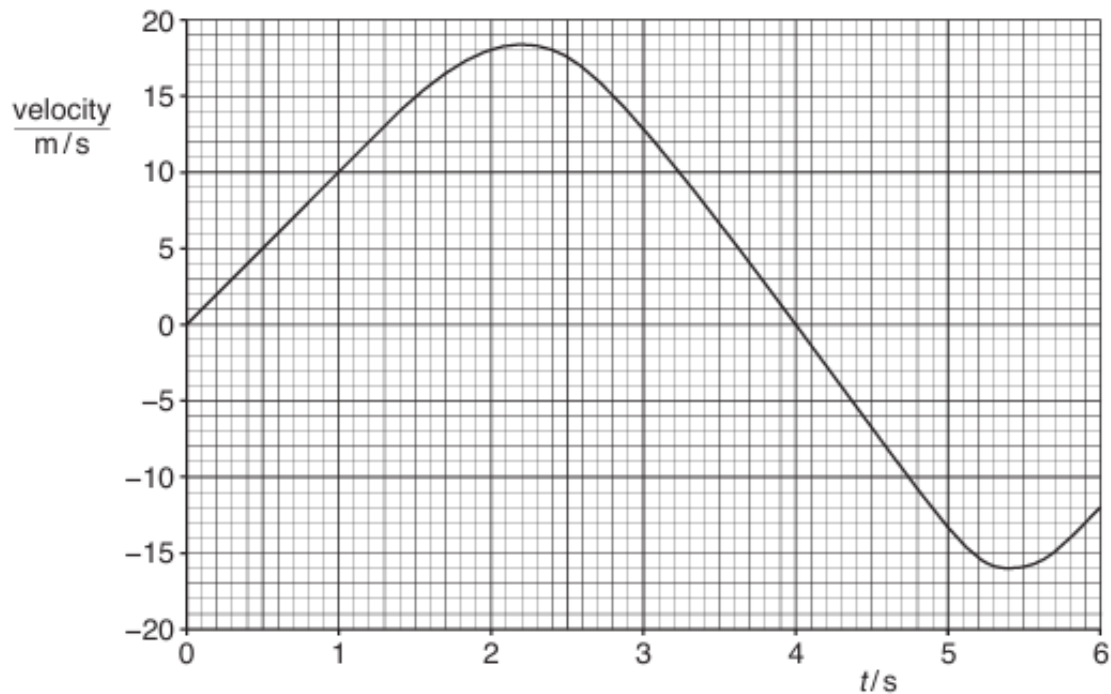


Fig. 11.2

(a) (i) State what is meant by *velocity*.

.....
..... [2]

(ii) State the difference between a positive velocity and a negative velocity.

.....
..... [1]

(iii) In the first 1.4 s the acceleration is uniform.

1. Using values from Fig. 11.2, determine the acceleration of the man in the first 1.4 s.

acceleration = [3]

2. Comment on your value of acceleration.

.....
..... [1]

22. O/N/14/22/Q9

(a) Explain what is meant by *uniform acceleration*.

.....

[2]

(b) Acceleration is a vector quantity.

State how a vector quantity differs from a scalar quantity.

.....
[1]

(c) A rock from space is travelling in a straight line at high speed when it enters the Earth's atmosphere. Fig. 9.1 is the speed-time graph for the rock from time $t = 0$ to time $t = 50$ s.

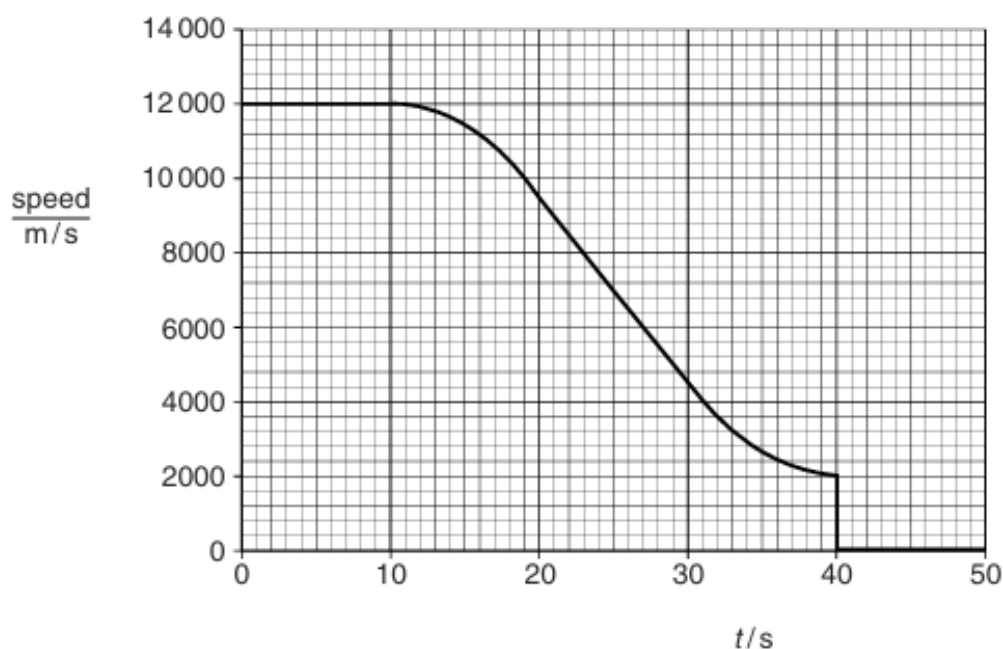


Fig. 9.1

(i) On Fig. 9.1, mark

- a letter X, where the rock is moving with a constant speed,
- a letter Y, where the rock has a uniform deceleration,
- a letter Z, where the rock has a non-uniform deceleration.

[3]

- (ii) At time $t = 25$ s, the mass of the rock is 8.4 kg. For the time $t = 25$ s, determine the size of the acceleration of the rock,

acceleration =[3]

23. M/J/14/22/Q9

A children's ride consists of a steel cable that runs between two posts of different heights, as shown in Fig. 9.1.

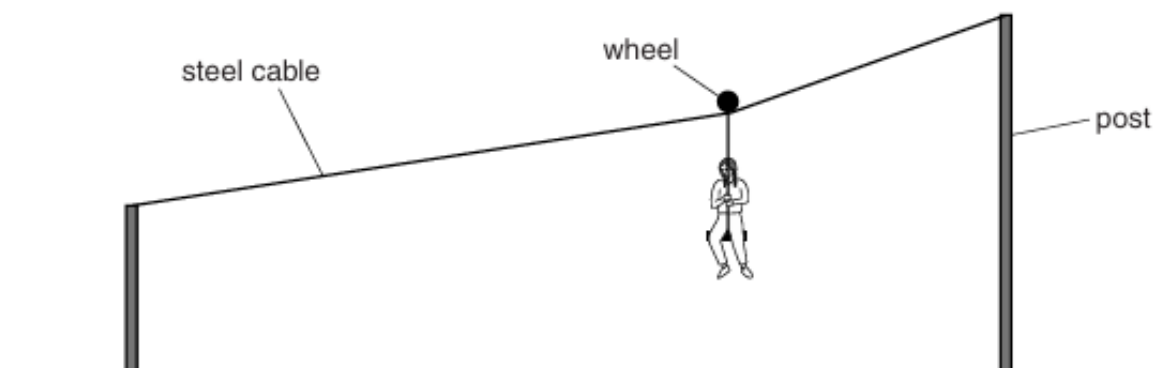


Fig. 9.1

A girl starts and finishes the ride at rest. Her horizontal motion can be taken as

- an initial uniform acceleration for 3.0 s, followed by
- a constant speed of 2.4 m/s for a further 5.0 s and
- a final uniform deceleration that lasts for 1.0 s.

(a) On Fig. 9.2, draw a speed-time graph of the horizontal motion.

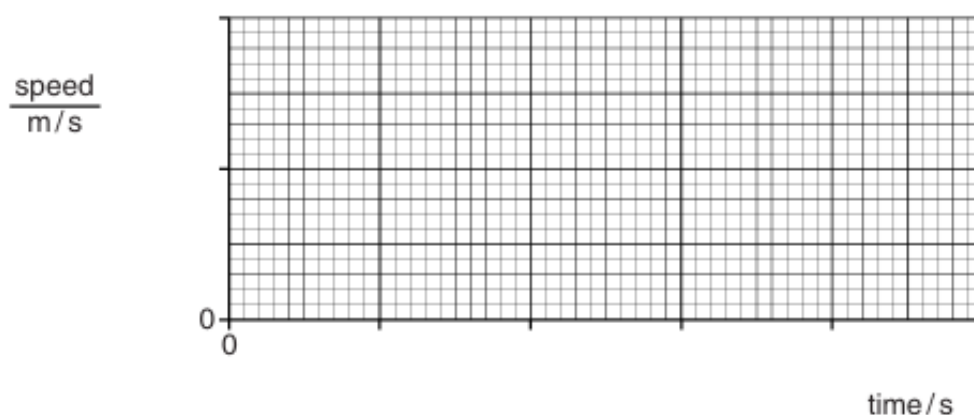


Fig. 9.2

[3]

(b) Explain what is meant by *uniform acceleration*.

.....

 [2]

- (c) The final deceleration is larger in size than the initial acceleration.

Explain how the data shows this.

.....
..... [1]

- (d) Calculate the horizontal distance travelled by the girl in the first 8.0 s.

distance = [3]

- (e) (i) The girl has a mass of 30 kg and falls a vertical distance of 1.6 m during the ride.

The gravitational field strength g is 10 N/kg.

Calculate the decrease in gravitational potential energy of the girl.

decrease in potential energy = [2]

- (ii) The gain in kinetic energy of the girl is less than the decrease in her potential energy. Suggest one reason for this.

.....
..... [1]

- (f) A group of pupils make measurements to show that the girl's speed is constant during the middle section of the ride.

Suggest what measurements are made and how they show that the speed is constant.

.....
.....
.....
.....
.....
..... [3]

24. O/N/13/22/Q10

Fig. 10.1 shows an electric train of mass $5.5 \times 10^5 \text{ kg}$.



Fig. 10.1

The train is initially at rest. The electric motor exerts a constant force and the train accelerates. Its acceleration decreases until the train reaches a speed of 40 m/s .

The train then continues at this constant speed.

- (a) (i)** On the axes in Fig. 10.2, sketch a speed-time graph for the train.

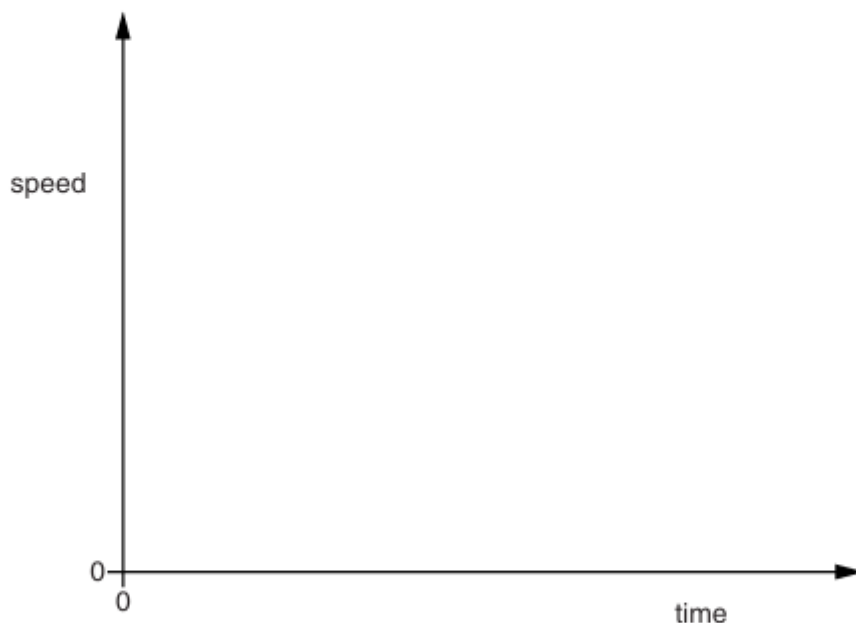


Fig. 10.2

[3]

- (ii)** Describe how the speed-time graph for the train is used to determine the distance travelled by the train when it is accelerating.

.....
.....
.....[2]

25. M/J/13/22/Q1

A ball rolls down a slope, as shown in Fig. 1.1.

The metre rule shows the position of the ball at times $t = 0$, 1.0 s, 2.0 s and 3.0 s.

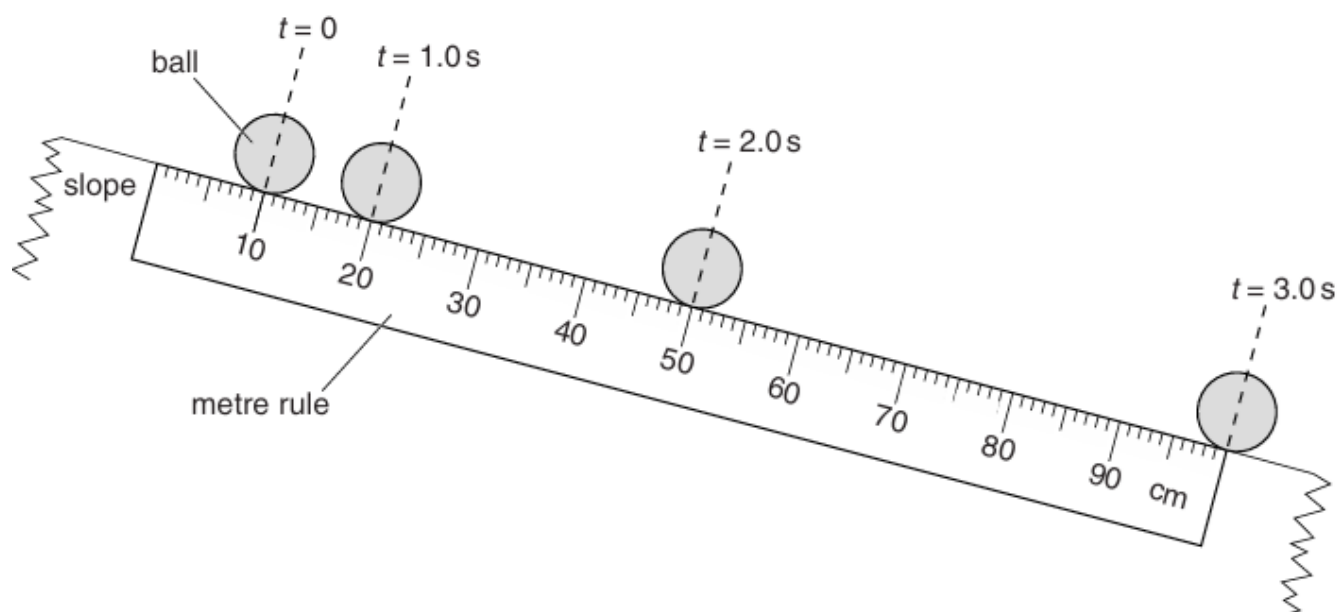


Fig. 1.1

(a) Explain how Fig. 1.1 shows that the ball is accelerating.

.....
..... [1]

(b) Calculate the average speed of the ball between $t = 1.0$ s and 3.0 s.

average speed = [2]

26. O/N/11/22/Q9

Fig. 9.1 is the speed-time graph for a racing car of total mass 650 kg as it sets off from rest at the start of a race.

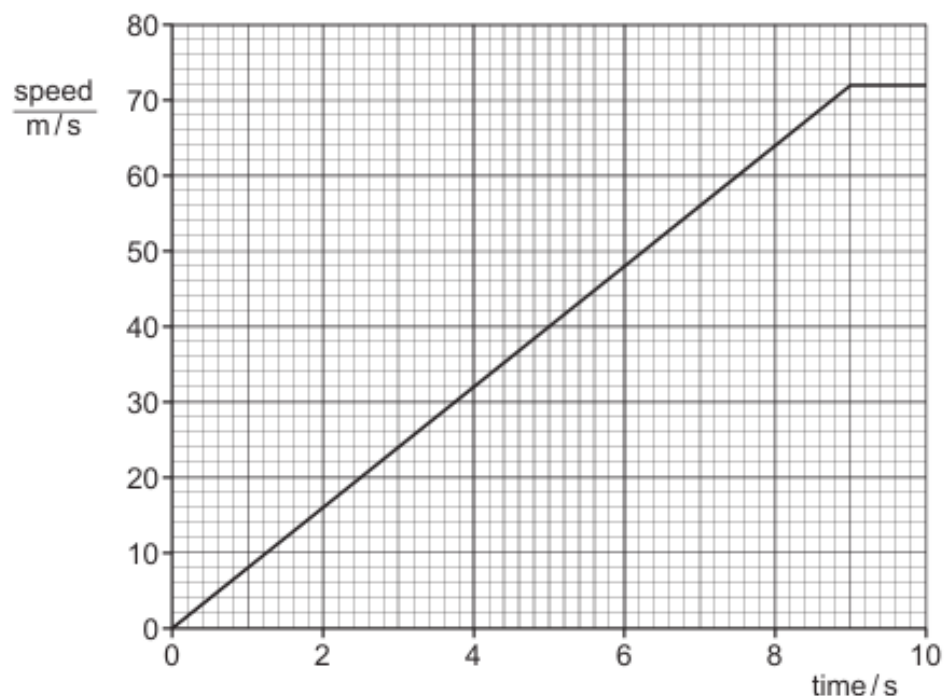


Fig. 9.1

The car travels in a straight line until time $t = 9.0$ s.

(a) State the speed of the car at time $t = 9.0$ s.

speed =[1]

(b) Calculate, for the car between times $t = 0$ and $t = 9.0$ s,

(i) the distance travelled,

distance =[2]

(ii) the acceleration of the car,

acceleration =[2]

27. M/J/11/22/Q9

Fig. 9.1 shows a sky-diver falling vertically.



Fig. 9.1

The sky-diver starts from rest at time $t = 0$.

His acceleration is non-uniform until he reaches a steady speed of 50 m/s at $t = 10$ s.

He opens his parachute at $t = 20$ s and decelerates until $t = 25$ s.

From $t = 25$ s he falls at a steady speed of 5 m/s.

(a) (i) On Fig. 9.2, draw the speed-time graph for the sky-diver.

[4]

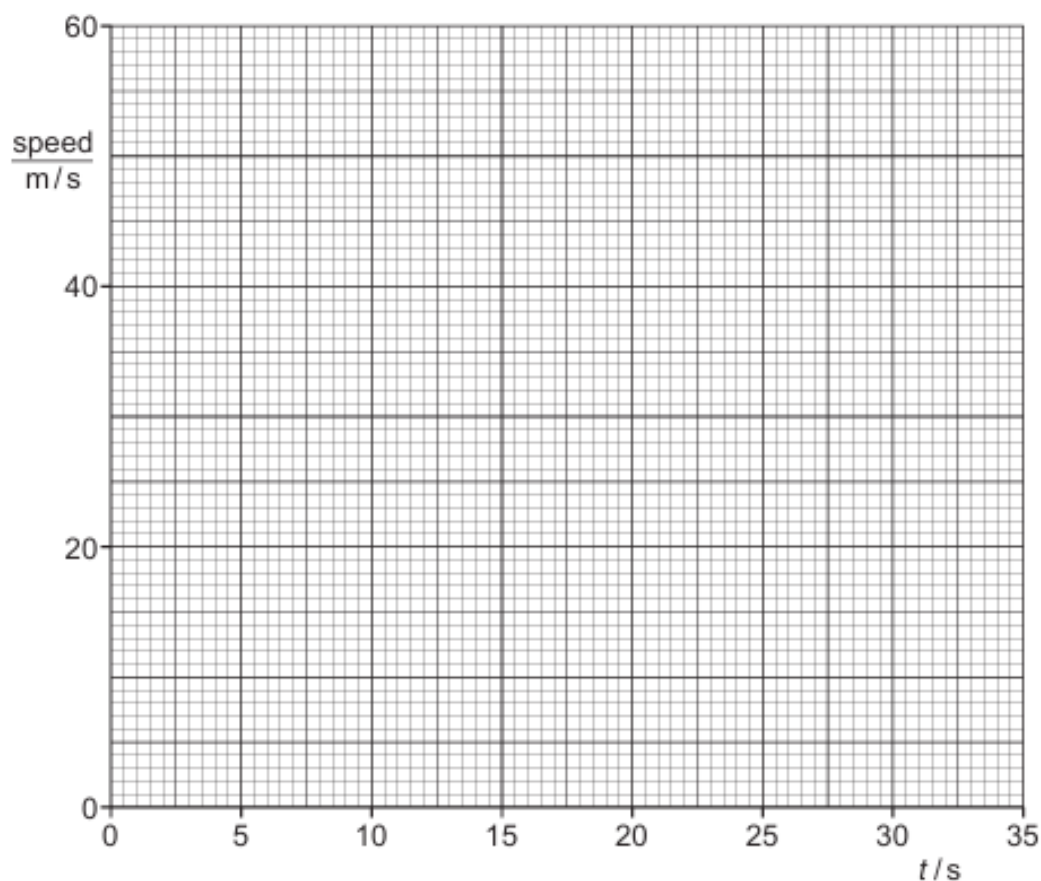


Fig. 9.2

(ii) State how your graph shows that the acceleration is non-uniform between $t = 0$ and $t = 10$ s.

.....
[1]

- (b) Explain, in detail, why **after** the sky-diver opens his parachute, he decelerates and eventually reaches a steady speed.

.....

.....

.....

.....

.....

.....[4]

- (c) For the time interval between $t = 10\text{ s}$ and $t = 20\text{ s}$, calculate the distance that the sky-diver falls.

distance =[1]

- (d) The mass of the sky-diver is 60 kg .

For the time interval between $t = 20\text{ s}$ and $t = 25\text{ s}$,

- (i) calculate the average deceleration of the sky-diver,

average deceleration =[2]

- (ii) calculate the average resultant force acting on the sky-diver,

force =[2]

- (iii) state how your graph in Fig. 9.2 may be used to obtain the distance that the sky-diver falls.

.....

.....[1]

28. M/J/11/21/Q1

Fig. 1.1 shows the speed-time graph for a car.

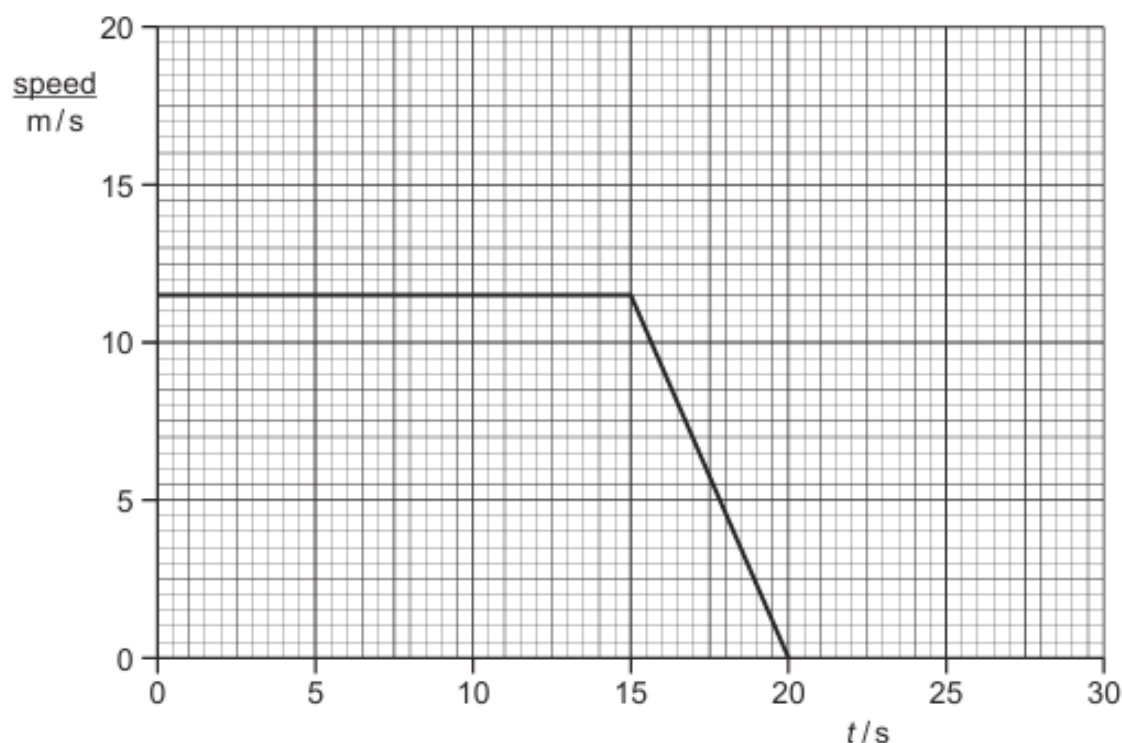


Fig. 1.1

At time $t = 15$ s, the brakes are applied and the car comes to rest with uniform deceleration.

- (a) (i) State the speed of the car just before the brakes are applied.

speed = [1]

- (ii) Explain what is meant by *uniform deceleration*.

.....

 [2]

- (b) A lorry travels at a constant speed of 18 m/s for 15 s. At time $t = 15$ s, the brakes are applied and the lorry slows down with the same deceleration as the car.

- (i) On Fig. 1.1, draw the speed-time graph for the lorry. [2]

- (ii) Explain how your graph shows that, while braking, the lorry travels further than the car.

.....
 [1]

29. O/N/10/22/Q1

Fig. 1.1 shows a cricket ball as it comes into contact with a cricket bat.



Fig. 1.1

The cricket ball has a mass of 0.16 kg and it hits the bat with a speed of 25 m/s . After being in contact with the bat for 0.0013 s , the ball rebounds with a speed of 22 m/s in the direction exactly opposite to its original direction.

(a) State the difference between *speed* and *velocity*.

.....
.....[1]

(b) Calculate

(i) the change in velocity of the cricket ball,

velocity change =[1]

(ii) the average acceleration of the ball whilst it is in contact with the bat,

acceleration =[2]

30. O/N/10/21/Q1

A window cleaner drops a sponge from the balcony of a hotel at time $t = 0$. Fig. 1.1 is the speed-time graph for the motion of the sponge.

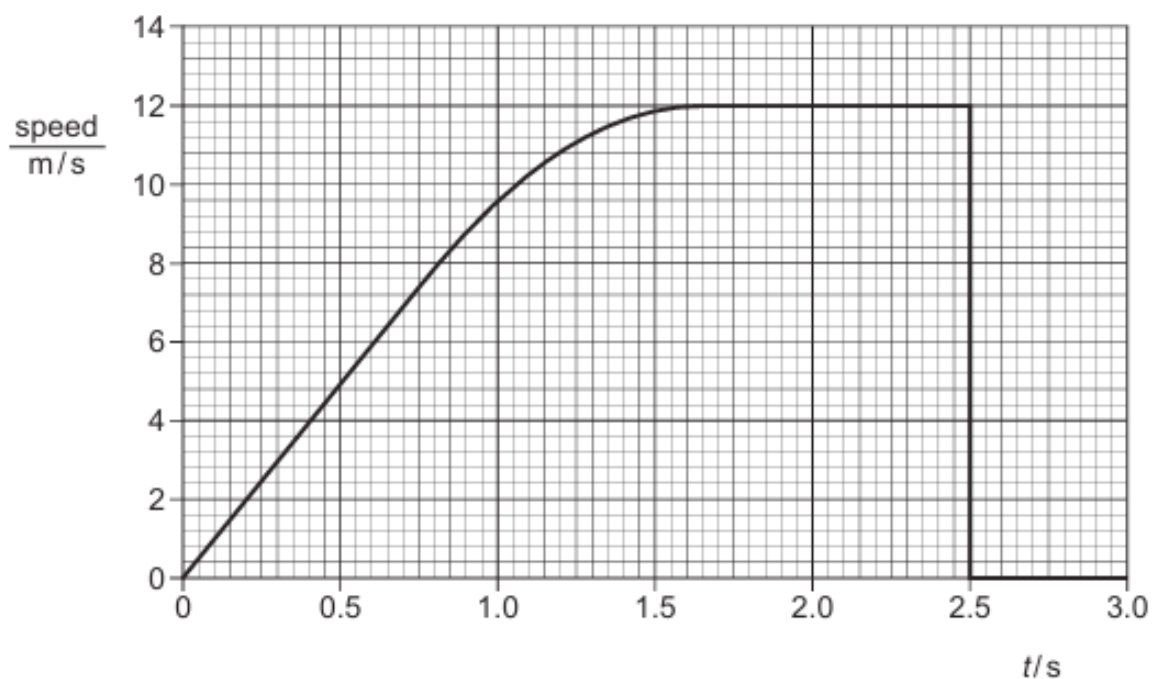


Fig. 1.1

(a) State a value for t when the sponge is

(i) moving with a uniform speed,

..... [1]

(ii) accelerating at a non-uniform rate,

..... [1]

(iii) decelerating.

..... [1]

(b) Calculate the distance travelled by the sponge between $t = 0$ and $t = 0.75$ s.

distance = [2]

31. M/J/09/Q1

A piece of paper falls from 4.0 m above the ground.

Fig. 1.1 shows how the height h above the ground varies with the time t .

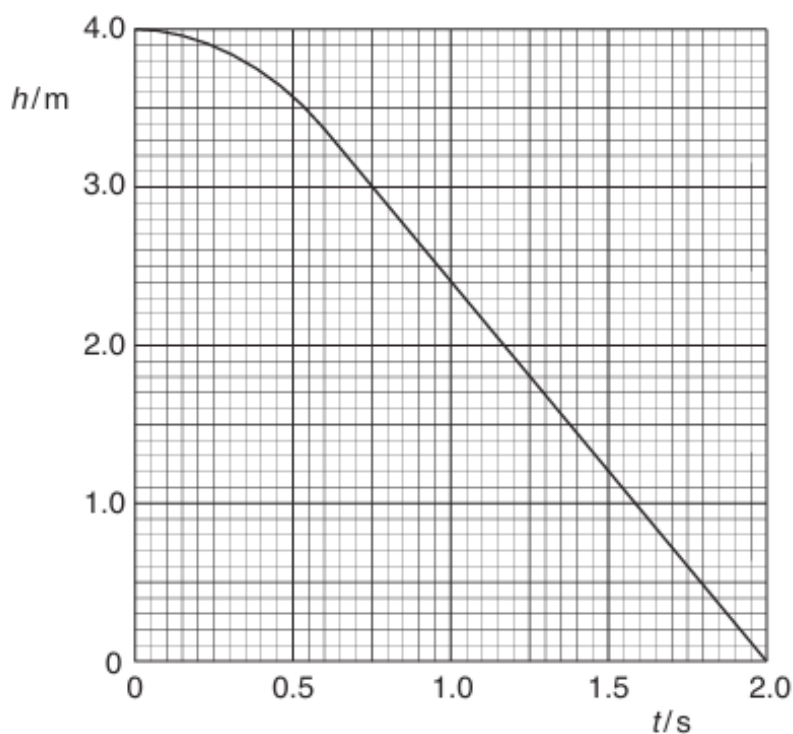


Fig. 1.1

- (a)** State what happens to the speed of the paper as it falls.

.....
.....
..... [2]

- (b)** Calculate the speed of the paper at time $t = 1.5$ s.

speed = [2]

32. O/N07/Q1

Fig. 1.1 is a graph of the vertical speed of the parachutist plotted against time t .

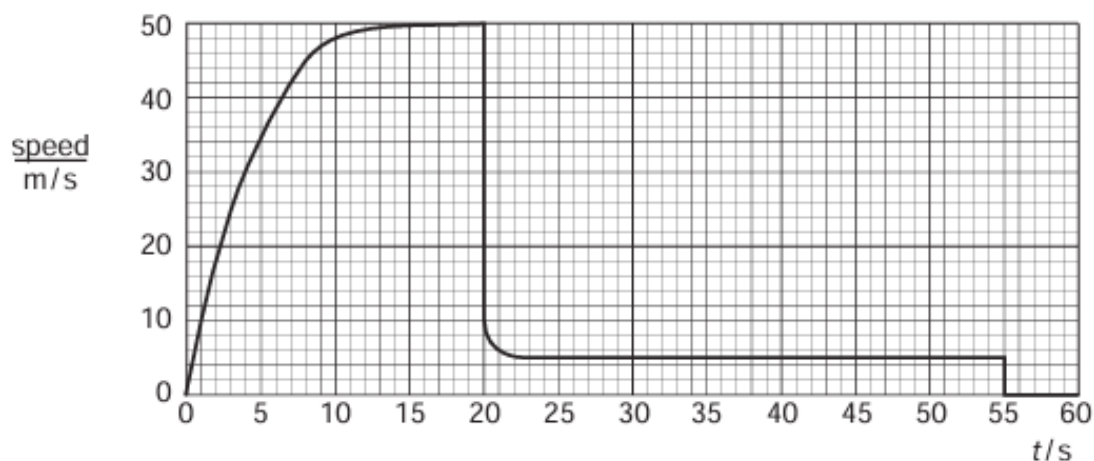


Fig. 1.1

- (a) State what happens at $t = 20$ s and $t = 55$ s.

at 20 s

at 55 s [1]

- (b) Describe the motion of the parachutist between $t = 0$ and $t = 20$ s.

.....

.....

..... [2]

- (c) Explain, in terms of the forces acting, why the speed of the parachutist is constant between $t = 25$ s and $t = 55$ s.

.....

.....

..... [2]

- (d) Calculate the distance travelled by the parachutist between $t = 25$ s and $t = 55$ s.

distance = [2]

33. M/J/07/Q1

Two athletes, A and B, run a 100 m race. At time $t = 0$, a gun is fired to start the race.

Fig. 1.1 shows the distance-time graph for the two athletes.

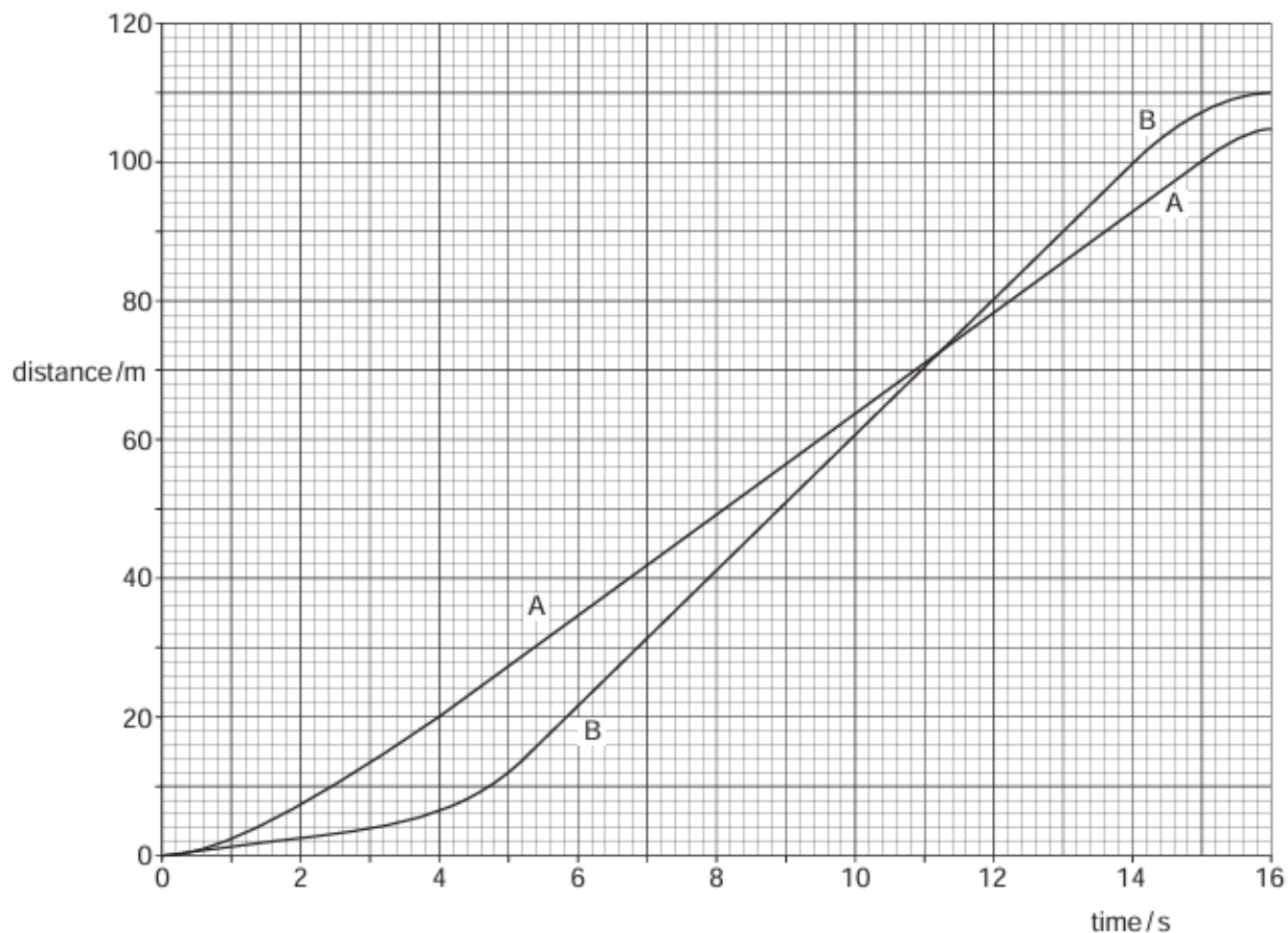


Fig. 1.1

(a) Describe the motion of athlete A during the first 8 s of the race.

.....

.....

.....

.....

..... [2]

(b) State the distance between the two athletes as the winner passes the 100 m mark.

..... [1]

(c) Calculate the speed of athlete A between $t = 4$ s and $t = 15$ s.

speed = [2]

34. O/N/06/Q1

Fig. 1.1 represents the motion of a car along a straight road. As the car approaches a small town, it slows down. The car travels at a constant speed from the start of the town to the end of the town. After passing through the town, the car speeds up.

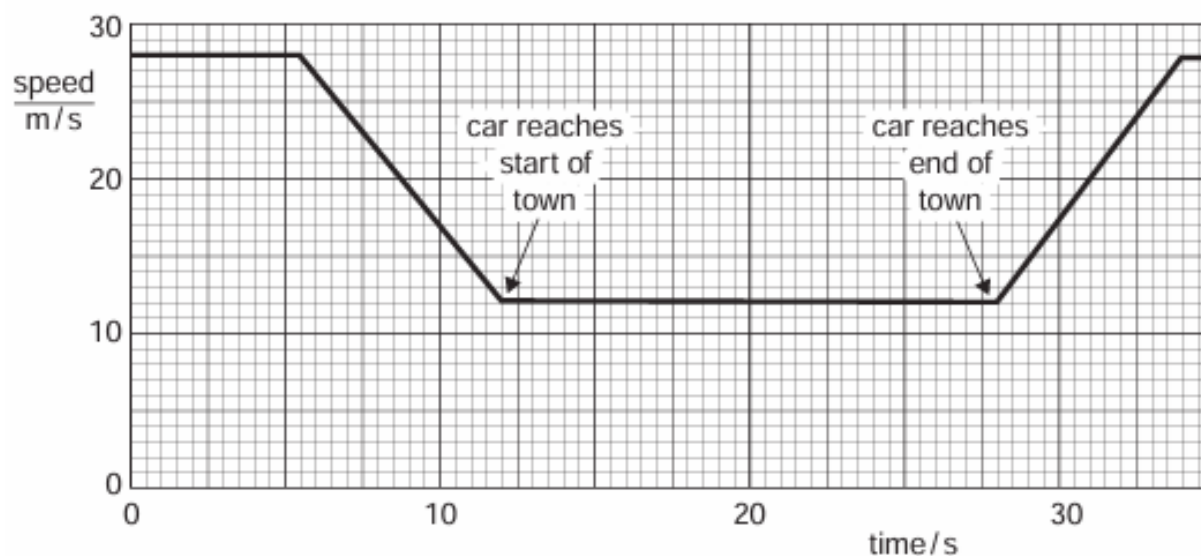


Fig. 1.1

- (a) (i) Determine the speed of the car in the town.

speed = [1]

- (ii) Determine the time taken by the car to pass through the town.

time taken = [1]

- (iii) Calculate the distance travelled by the car in the town.

distance = [1]

- (b) The car accelerates after passing through the town.

Calculate the acceleration. Give the unit of your answer.

acceleration = [3]

35. M/J/06/Q2

A cyclist starts from rest. He accelerates and then travels at a constant speed. At 12 s, the cyclist applies the brakes and slows down. Photographs are taken of the cyclist at 4 s intervals. Fig. 2.1 shows the results.

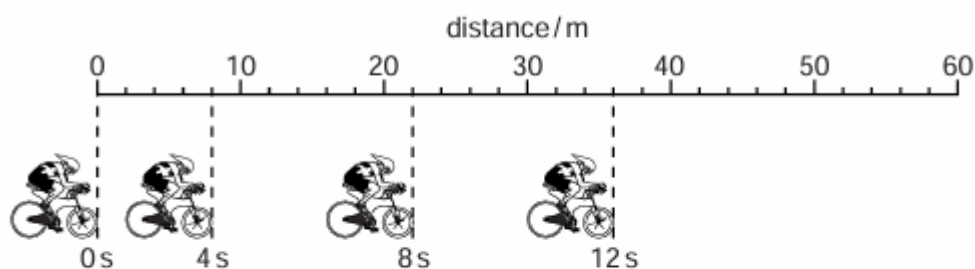


Fig. 2.1

- (a) On Fig. 2.1, draw a possible position of the front wheel of the cycle at 16 s. [1]
- (b) On Fig. 2.2, plot a distance-time graph of the cyclist for the first 16 s. [4]

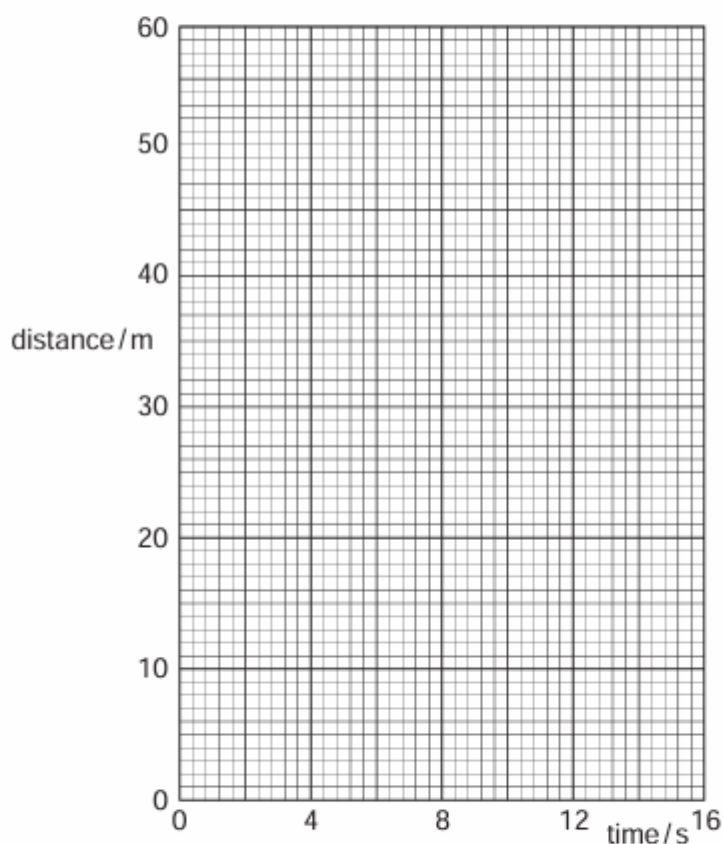


Fig. 2.2

- (c) Calculate the average speed of the cyclist during the first 12 s.

average speed = [2]

36. O/N/04/Q1

Fig. 1.1 shows a simplified speed-time graph for a train that travels between two stations.

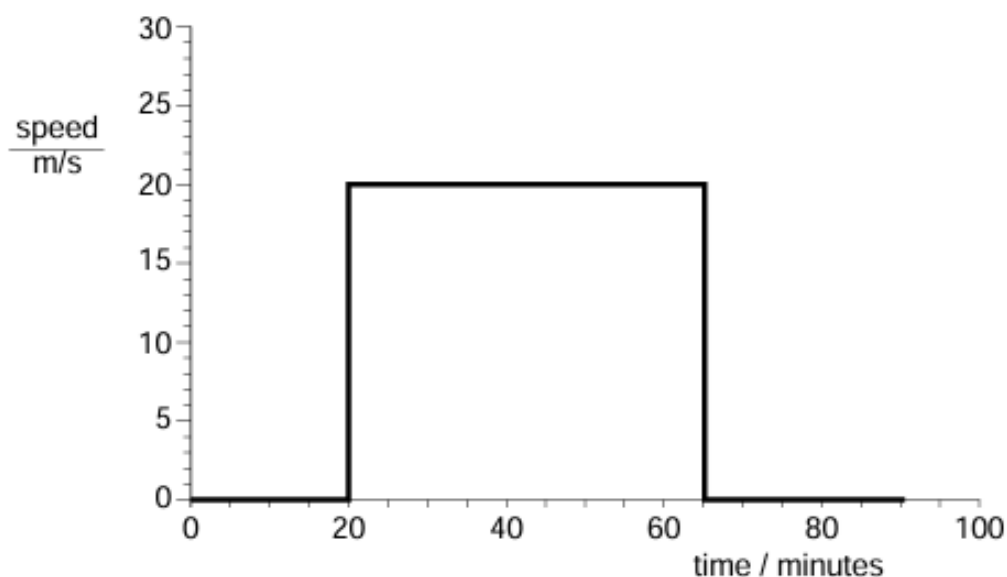


Fig. 1.1

- (a)** Describe the motion of the train.

.....

.....

.....

.....

.....[2]

- (b)** Calculate the distance travelled by the train between the two stations.

distance = [2]

- (c)** Another train travels between the same two stations on a parallel track. This train travels at a constant speed. It starts its journey at time $t = 0$ and finishes at $t = 90$ minutes.

On Fig. 1.1, draw the speed-time graph for this train. [2]

37. M/J/04/Q4

Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

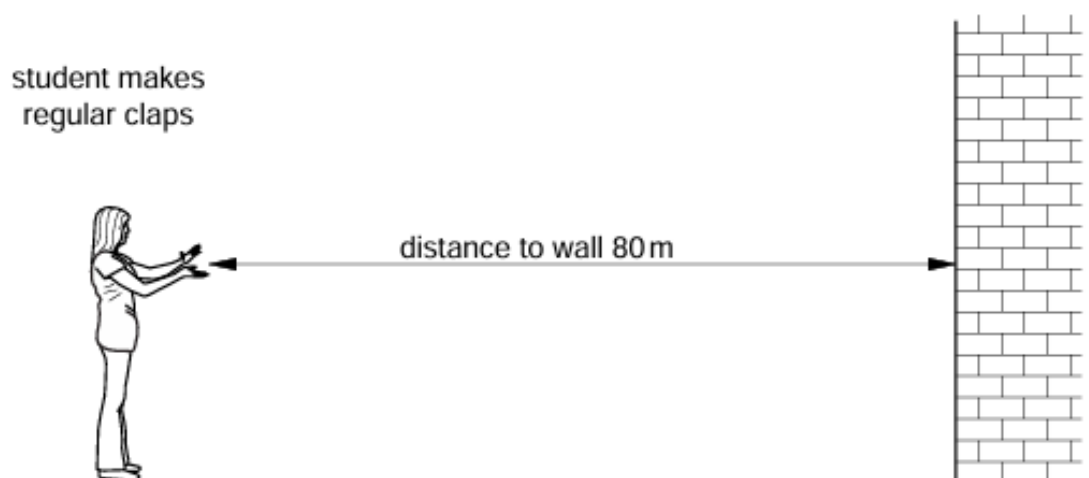


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....
.....[1]

(b) Calculate the time between claps.

time = [3]

(c) Calculate the number of claps in 1 minute.

number of claps = [2]